19. In fuel cell H_2 and O_2 react to produce electricity. In the process, H_2 gas is oxidised at the anode and O_2 gas is reduced at cathode. If 44.8 litre of H_2 at 25^oC and 1atm pressure reacts in 10 minutes, what is average current produced? If the entire current is used for electro deposition of Cu from Cu^{2+}, how many grams of Cu deposited?

20. The same amount of electricity was passed through two separate electrolytic cells containing solutions of nickel nitrate and chromium nitrate respectively. If 2.935g of Ni was deposited in the first cell. The amount of Cr deposited in the another cell? Give : molar mass of Nickel and chromium are 58.74 and $52gm^{-1}$ respectively.

21. A copper electrode is dipped in 0.1M copper sulphate solution at 25^oC. Calculate the electrode potential of copper. [Given: $E^\circ_{Cu^{2+}|Cu} = 0.34\ V$].

22. For the cell $Mg\,(s)\left|Mg^{2+}(aq)\right\|Ag^{+}(aq)\left|Ag\,(s)\right.$, calculate the equilibrium constant at 25^oC and maximum work that can be obtained during operation of cell. Given : $E^\circ_{Mg^{2+}|\,Mg} = -2.37V$ and $E^\circ_{Ag^{+}|Ag} = 0.80V$.

23. 9.2×10^{12} litres of water is available in a lake. A power reactor using the electrolysis of water in the lake produces electricity at the rate of $2\times10^6\ Cs^{-1}$ at an appropriate voltage. How many years would at take to completely electrolyse the water in the lake. Assume that there is no loss of water except due to electrolysis.

24. Derive an expression for Nernst equation

25. Write a note on sacrificial protection.

26. Explain the function of H_2 - O_2 fuel cell.

27. Ionic conductance at infinite dilution of Al^{3+} and SO_4^{2-} are 189 and 160 mho cm^2 $equiv^{-1}$. Calculate the equivalent and molar conductance of the electrolyte $Al_2(SO_4)_3$ at infinite dilution.

XII U9 Electro Chemistry.indd 67 11-01-2021 14:44:51